

1

A stock price is currently \$40. It is known that at the end of one month it will be either \$42 or \$38. The risk-free interest rate is 8% per annum with continuous compounding. What is the value of a one-month European call option with a strike price of \$39?

Following the way that we set up the portfolios in the text, we consider a portfolio with Δ shares of a stock and short 1 call option with the specification as in the problem. The portfolio value would be the following in each situation:

- If the price becomes \$42: $42\Delta - 3$
- If the price becomes \$38: 38Δ

As the portfolio is riskless when these are equal, we have

$$42\Delta - 3 = 38\Delta$$

$$4\Delta = 3$$

$$\Delta = 0.75$$

So the riskless portfolio is longing 0.75 shares and shorting 1 call option. The current value of the portfolio is therefore $38 \cdot 0.75 = 28.5$, so with discounting at the risk-free rate over a month, we have

$$(0.75 \cdot 40 - f)e^{0.08 \cdot \frac{1}{12}} = 28.5$$

$$(30 - f)e^{\frac{1}{150}} = 28.5$$

$$30 - f = 28.3106319284$$

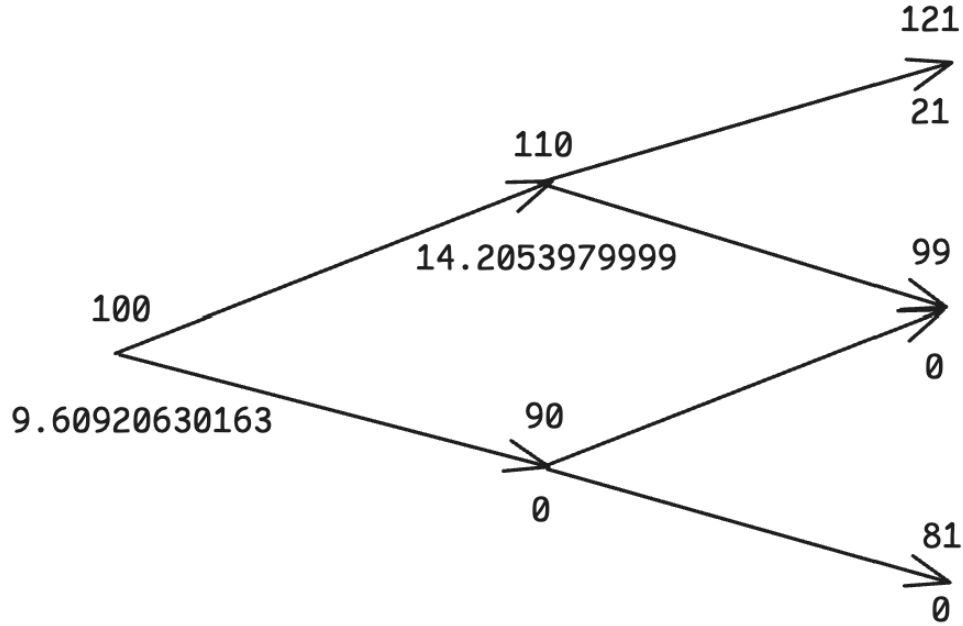
$$f = 1.6893680716 \approx \boxed{\$1.69}$$

2

A stock price is currently \$100. Over each of the next two six-month periods it is expected to go up by 10% or down by 10%. The risk-free interest rate is 8% per

annum with continuous compounding. What is the value of a one-year European call option with a strike price of \$100?

In this case, we have $u = 1.10, d = 0.90, \Delta t = 0.5, r = 0.08$. We set up the tree like so:



We know that

$$p = \frac{e^{r\Delta t} - d}{u - d}$$

$$p = \frac{e^{0.08 \cdot 0.5} - 0.9}{1.10 - 0.90}$$

$$p = \frac{e^{0.04} - 0.9}{0.2}$$

$$p = \frac{0.14081077419}{0.2} = 0.70405387095$$

We use this to work backwards in the tree. First step:

$$f_u = e^{-0.08 \cdot 0.5} \cdot ((0.70405387095 \cdot 21) + ((1 - 0.70405387095) \cdot 0))$$

$$= e^{-0.04} \cdot 14.78513129 = 14.2053979999$$

And last step to the beginning:

$$f = e^{-0.08 \cdot 0.5} \cdot ((0.70405387095 \cdot 14.2053979999) + ((1 - 0.70405387095) \cdot 0))$$

$$= e^{-0.04} \cdot 10.0013654502 = 9.60920630163 \approx \boxed{\$9.61}$$

3

For the situation considered in Problem 2, what is the value of a one-year European put option with a strike price of \$100? Verify that the European call and European put prices satisfy put-call parity.

With the same setup as above, we can recreate the tree or just calculate the option value directly:

$$\begin{aligned} f &= e^{-r\Delta t}[pf_u + (1-p)f_d] \\ &= e^{-2r\Delta t}[p^2f_{uu} + 2p(1-p)f_{ud} + (1-p)^2f_{dd}] \\ &= e^{-2 \cdot 0.08 \cdot 0.5}[0.70405387095 \cdot 0 + 2 \cdot 0.70405387095 \cdot (1 - 0.70405387095) \cdot 1 + (1 - 0.70405387095)^2 \cdot 19] \\ &= e^{-0.08}[2 \cdot 0.70405387095 \cdot 0.29594612905 + 0.29594612905^2 \cdot 19] \\ &= 0.92311634638 \cdot (0.4167240355 + 1.66409811451) \\ &= 1.92084094058 \approx \boxed{\$1.92} \end{aligned}$$

Put-call parity tells us that

$$\begin{aligned} c + Ke^{-rT} &= p + S_0 \\ 9.61 + 100e^{-0.08 \cdot 1} &= 1.92 + 100 \\ 9.61 + 92.311634638 &= 101.92 \\ 101.921634638 &\approx 101.92 = 101.92 \end{aligned}$$

which indeed shows us that put-call parity does hold!

4

A stock price is currently \$40. It is known that at the end of three months it will be either \$45 or \$35. The risk-free rate of interest with quarterly compounding is 8%

per annum. Calculate the value of a three-month European put option on the stock with an exercise price of \$40. Verify that no-arbitrage arguments and risk-neutral valuation arguments give the same answers.

We set up a portfolio with Δ shares and 1 option, which works in this case since we are working with a put option, which is essentially negative. So we know that the value of the portfolio would be the following in each situation:

- If the price becomes \$45: 45Δ
- If the price becomes \$35: $35\Delta - 5$

As the portfolio is riskless when these are equal, we have

$$45\Delta = 35\Delta - 5$$

$$10\Delta = -5$$

$$\Delta = -0.5$$

So the riskless portfolio is essentially short 0.5 shares and longing 1 put option. We see that the value of the portfolio is $45 \cdot -0.5 = -22.5$. Discounting at the risk-free rate over the 3 months, we can solve for f accordingly:

$$(-0.5 \cdot 40 + f) \cdot (1 + 0.08 \cdot \frac{3}{12}) = -22.5$$

$$(-20 + f) \cdot 1.02 = -22.5$$

$$-20 + f = -22.0588235294$$

$$f = 2.0588235294 \approx \boxed{\$2.06}$$

We confirm this using the risk-neutral valuation argument:

$$45p + 35(1 - p) = 40 \cdot (1 + 0.08 \cdot \frac{3}{12})$$

$$45p + 35 - 35p = 40 \cdot 1.02$$

$$10p = 40.8 - 35$$

$$p = 0.58$$

So the expected value of the option is

$$\begin{aligned} & 0.58 \cdot 0 + (1 - 0.58) \cdot 5 \\ & = 0 + 0.42 \cdot 5 = 2.10 \end{aligned}$$

and adjusting for present value is $2.10 \div 1.02 = 2.05882352941 \approx 2.06$, which is consistent with what we previously found using the no-arbitrage argument.

5

Consider a European call option on a non-dividend-paying stock where the stock price is \$40, the strike price is \$40, the risk-free rate is 4% per annum, the volatility is 30% per annum, and the time to maturity is six months.

(a)

Calculate u , d , and p for a two step tree

We do a two step tree over six months, so we have $\Delta t = 0.5 \div 2 = 0.25$. We therefore have

$$u = e^{0.3 \cdot \sqrt{0.25}} = e^{0.15} = \boxed{1.16183424273}$$

and

$$d = \frac{1}{u} = \frac{1}{1.16183424273} = \boxed{0.86070797642}$$

Finally:

$$\begin{aligned} p &= \frac{e^{r\Delta t} - d}{u - d} \\ &= \frac{e^{0.04 \cdot 0.25} - 0.86070797642}{1.16183424273 - 0.86070797642} \\ p &= \frac{e^{0.01} - 0.86070797642}{0.30112626631} = \frac{0.14934219066}{0.30112626631} = \boxed{0.49594541349} \end{aligned}$$

(b)

Value the option using a two step tree. We fill in the tree using the values we found in part *a*, with the prices on the tree as follows:

- $S_u = 40u = 40 \cdot 1.16183424273 = 46.4733697092$
- $S_{uu} = 40u^2 = 40 \cdot 1.16183424273^2 = 53.9943523032$
- $S_{ud} = S_{du} = 40du = 40 \cdot 1.16183424273 \cdot 0.86070797642 = 39.9999999998 \approx 40$
- $S_d = 40d = 40 \cdot 0.86070797642 = 34.4283190568$
- $S_{dd} = 40d^2 = 40 \cdot 0.86070797642^2 = 29.6327288268$

And the corresponding f values:

$$\begin{aligned} f_u &= e^{-0.04 \cdot \frac{1}{2} \div 2} \cdot [(0.49594541349 \cdot 13.9943523032) + (1 - 0.49594541349) \cdot 0] \\ &= e^{-0.01} \cdot (0.49594541349 \cdot 13.9943523032) \\ f_u &= 0.99004983374 \cdot 6.94043483954 = 6.87137635897 \end{aligned}$$

and

$$\begin{aligned} f &= e^{-0.04 \cdot \frac{1}{2} \div 2} \cdot [(0.49594541349 \cdot 6.87137635897) + (1 - 0.49594541349) \cdot 0] \\ f &= 0.99004983374 \cdot 3.40782758959 = 3.37391913849 \approx \boxed{\$3.37} \end{aligned}$$

(c)

Verify that DerivaGem gives the same answer.

At each node:

Upper value = Underlying Asset Price

Lower value = Option Price

Values in red are a result of early exercise.

Strike price = 40

Discount factor per step = 0.9900

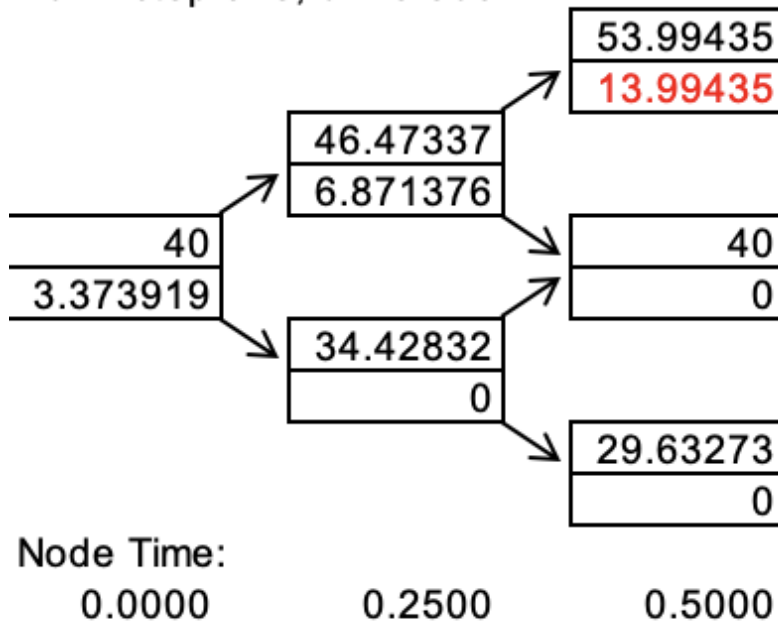
Time step, $dt = 0.2500$ years, 91.25 days

Growth factor per step, $a = 1.0101$

Probability of up move, $p = 0.4959$

Up step size, $u = 1.1618$

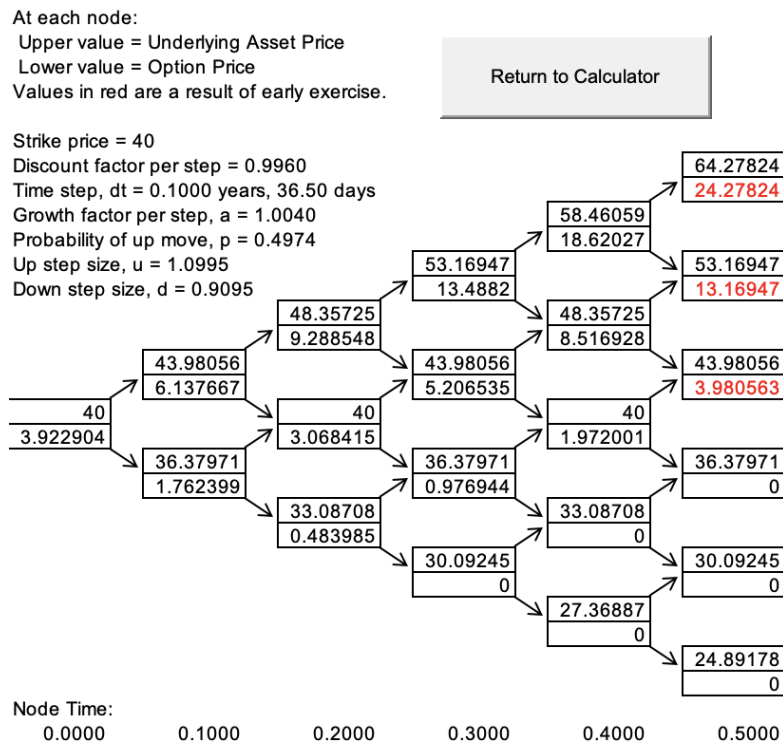
Down step size, $d = 0.8607$



Looking at the DerivaGem tree, we do indeed see that we have the same price for the option.

(d)

Use DerivaGem to value the option with 5, 50, 100, and 500 time steps.



We have that the price with 5 steps is approximately **\$3.92**.

Underlying Type:
Equity

Stock Price: 40.00
Volatility (% per year): 30.00%
Risk-Free Rate (% per year): 4.00%

Option Type:
Binomial: European

Life (Years): 0.5000
Strike Price: 40.00
Tree Steps: 50

☐ Implied Volatility
☐ Put
☒ Call

Calculate **Display Tree**

Results:
Price: 3.73942017
Delta (per \$): 0.57900437
Gamma (per \$ per \$): 0.04679706
Vega (per %): 0.23219381
Theta (per day): -0.01136356
Rho (per %): 0.09777571

We have that the price with 50 steps is approximately **\$3.74**.

Underlying Type:

Equity

Stock Price:

40.00

Volatility (% per year):

30.00%

Risk-Free Rate (% per year):

4.00%

Time (Yrs)

Dividend

Option Type:

Binomial: European

Life (Years):

0.5000

Strike Price:

40.00

Tree Steps:

100

☐ Imply Volatility

☐ Put
 ☒ Call

Calculate

Display Tree

Results:

Price:

3.73942017

Delta (per \$):

0.57900437

Gamma (per \$ per \$):

0.04679706

Vega (per %):

0.23219381

Theta (per day):

-0.01136356

Rho (per %):

0.09777571

We have that the price with 100 steps is also approximately **\$3.74**.

Underlying Type:

Equity

Stock Price:

40.00

Volatility (% per year):

30.00%

Risk-Free Rate (% per year):

4.00%

Time (Yrs)

Dividend

Option Type:

Binomial: European

☐ Imply Volatility

☐ Put
 ☒ Call

Life (Years):

0.5000

Strike Price:

40.00

Tree Steps:

500

Calculate

Display Tree

Results:

Price:

3.73942017

Delta (per \$):

0.57900437

Gamma (per \$ per \$):

0.04679706

Vega (per %):

0.23219381

Theta (per day):

-0.01136356

Rho (per %):

0.09777571

We have that the price with 500 steps is also approximately **\$3.74**.